

The Eikonal Bridge - Brief Guide

A Brief Guide to the Unified Architecture

From Classical Lens Design to Quantum Photonics

Version History:

- **v0 (Mar 4, 2026)**: Early LinkedIn post: “On My Upcoming Physics Colloquium Talk at NCU (posted date: March 4, 2026)”; thoughts on worked examples from basic to advanced.
- **v19 (Apr 18, 2026)**: Preliminary technical document.

Abstract

The Eikonal Bridge is one identity, $S = F$, linking the classical Strehl ratio to the quantum state fidelity. This brief guide presents its unified architecture in three strokes: **(i) one Master Loss Functional** that generates every practical figure of merit by parameter specialization; **(ii) six derivations D1–D6** arranged both as a reduction lattice (structural view) and as an efficiency plane (computational view), with D1 at the hub; and **(iii) eleven problem classes PC1–PC11** grouped by bridge-status, spanning the design space from a lossless imaging pupil to a fully multi-mode CPTP channel. Five figures and one guide-rules table (landscape) exhibit the whole framework at a glance; a final section treats the coverage matrix’s empty cells as explicit open problems. The document is self-contained: every symbol, acronym, and reference is defined or cited within these pages.

Reading path — three ways to use this guide

- **Executive (15 min)**: Read §1, §2, and skim the four figures (§3, §5, §6, §7).
- **Practitioner (45 min)**: Add §4 (problem classes) and §8 (the 10-step decision procedure). Print Table 3 (landscape) as a reference card.
- **Full read (2 hr)**: Include §3 (six derivations) for the rigor behind $S = F$, and §6.1 for the open-problems map.

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List of Abbreviations

Acronym	Expansion (one-sentence definition)
AO	Adaptive optics — active wavefront correction using deformable mirrors.
CODE V	Commercial ray-tracing and optical design suite (Synopsys Inc.).
CPO	Co-packaged optics — optical I/O integrated with silicon electronics.
CPTP	Completely positive trace-preserving map — the general quantum channel form.
CRLB	Cramér–Rao lower bound — variance floor for unbiased parameter estimators.
CV-QKD	Continuous-variable quantum key distribution.
DEE	Differentiable Eikonal Engine — the JAX architecture proposed in the book.
FMF	Few-mode fiber — an optical fiber supporting 2–10 guided modes.
JAX	Google’s automatic-differentiation numerical library.
LP	Linearly-polarized mode family of step-index fibers (LP_{01} , LP_{11} , ...).
MTF	Modulation transfer function — spatial-frequency response of an imaging system.
NA	Numerical aperture — sin of the marginal-ray angle at the image.
OAM	Orbital angular momentum of light.
OPD	Optical path difference — deviation of a real wavefront from the ideal.
PC	Problem class — one of the eleven entries in the book’s atlas.
PDF	Probability density function.
POVM	Positive operator-valued measure — the general quantum measurement form.
PSF	Point spread function — image of a point source.
PSD	Power spectral density — Fourier transform of an autocorrelation.
QFI	Quantum Fisher information.
QKD	Quantum key distribution.
RMS	Root-mean-square.
RSoft	Commercial photonic-device simulation suite (Synopsys Inc.).
SLD	Symmetric logarithmic derivative — generator of the QFI.
SMF	Single-mode fiber.
SPDC	Spontaneous parametric down-conversion — common heralded-photon source.
SPIE	International Society for Optics and Photonics.
WFE	Wavefront error.

List of Symbols

Symbol	Meaning (with units where relevant)
λ	Wavelength of light (m or nm).
$\mathcal{A}$	Pupil aperture (2D region).
A	Area of $\mathcal{A}$ (m ²).
$u_0(\mathbf{r})$	Ideal pupil mode, normalized by $\int u_0 ^2 = 1$.
$u_1(\mathbf{r})$	Aberrated mode $u_1 = u_0 e^{i\Delta\phi}$.
$\Delta W(\mathbf{r})$	Wavefront error at pupil position $\mathbf{r}$ (m).
$\Delta\phi(\mathbf{r})$	Phase aberration $(2\pi/\lambda) \Delta W$ (rad).
σ_W	RMS wavefront error (m).
σ_ϕ	RMS phase aberration (rad); $\sigma_\phi = (2\pi/\lambda)\sigma_W$.
γ_2	Excess kurtosis of the phase distribution.
S	Strehl ratio (classical).
F	Quantum state fidelity.
$\mathcal{L}$	Master loss functional [Eq. (2)].
$\hat{\rho}_{\text{tgt}}$	Target quantum state (density operator).
K_k	Kraus operator of a CPTP channel; $\sum_k K_k^\dagger K_k = I$.
ξ	Tightening factor, classical-to-quantum spec ratio.
c_0	Survival amplitude $c_0 = \langle u_0 u_1 \rangle$.
η	Coupling (or bucket) efficiency, classical sense.
$W(q, p)$	Wigner quasi-probability distribution (phase-space).
$\mathcal{F}_Q$	Quantum Fisher information (rad ⁻²).
Δ_{bridge}	Bridge gap $\eta - F$ (detector-mode mismatch).
$\mathcal{B}$	Bridge-break factor (PC3 diagnostic, unitless, ≤ 1).
τ	Coherence ratio ℓ_c/ℓ_ϕ (PC3 input).
$\rho_{A,\phi}$	Amplitude–phase correlation coefficient (PC10 diagnostic).
Y	Yield $P(F > F_{\text{spec}})$.
$\mathcal{E}(\hat{\rho})$	Action of a CPTP channel on density operator $\hat{\rho}$.

1 The Bridge Identity

Why this identity matters

Classical lens design and quantum photonics speak two different dialects. Lens designers work in terms of the Strehl ratio S , the wavefront error σ_W , and Zernike coefficients extracted from ray traces in simulation tools, e.g., CODE V or Zemax [1, 2]. Quantum photonicists work in terms of state fidelity F , mode overlap, and Kraus operators describing decoherence [14]. When a quantum photonicist asks a lens designer for specifications on a photon-delivery optic, these two vocabularies force a painful translation: does an $S = 0.9$ imaging system support a 99%-fidelity quantum gate? Does a vignetted pupil equal a 5% amplitude-damping channel? For decades the translation was done ad hoc, with different groups reinventing different conversions for different applications.

The identity

The translation collapses into one identity. For a scalar paraxial single-photon state with pupil mode $u_0(\mathbf{r})$ on aperture $\mathcal{A}$ of area A , aberrated by a unitary phase $\Delta\phi(\mathbf{r}) = (2\pi/\lambda) \Delta W(\mathbf{r})$, define the aberrated mode $u_1 = u_0 e^{i\Delta\phi}$. Then:

$$S = \left| \frac{1}{A} \int_{\mathcal{A}} e^{i\Delta\phi(\mathbf{r})} d^2r \right|^2 = F = |\langle u_0 | u_1 \rangle|^2. \quad (1)$$

Compact proof sketch. With uniform amplitude $u_0 = A^{-1/2} \mathbf{1}_{\mathcal{A}}$, normalization $\int |u_0|^2 = 1$ is automatic. Then the overlap $\langle u_0 | u_1 \rangle = (1/A) \int_{\mathcal{A}} e^{i\Delta\phi} d^2r$ is exactly the pupil average that defines S . Taking modulus-squared of both sides gives Eq. (1). No approximation is involved within the stated hypotheses (scalar, paraxial, pure state, phase-only aberration, uniform pupil amplitude).

Engineering consequence

Whatever Strehl ratio a classical lens design achieves, a single-mode photon passing through that design inherits the same value as its quantum state fidelity. Classical specifications translate directly to quantum specifications, with a typical tightening factor $\xi \approx 4.5\times$ on σ_W when moving from $S > 0.8$ (classical imaging criterion) to $F > 0.99$ (quantum gate threshold). That is, an imaging-grade lens with $\sigma_W \approx \lambda/14$ must be reworked to $\sigma_W \approx \lambda/63$ for a gate-fidelity target. **This is the book's single most practical deliverable: an existing CODE V design can be re-specified for quantum use without redesign; one scalar multiplication on its wavefront budget suffices.** Here the book is "Eikonal Bridge." (see <https://github.com/jlchern-251016>)

The remaining sections build out the full architecture: one master functional that generates every such translation (§2); six derivations, arranged as a reduction lattice (§3); eleven problem classes grouped by bridge-status (§4); four figures displaying the architecture at a glance (§3, §5–§7); a ten-step decision procedure (§8); and an open-problems map derived from the coverage matrix's empty cells (§6.1).

2 The Master Loss Functional

The bridge identity of Eq. (1) applies to the simplest case: lossless, single-mode, unitary phase, static aberration. Real optical systems depart from this corner along many axes — amplitude loss, dephasing, multi-mode scattering, high-NA vector effects. A single unifying expression accommodates all of these by specialization:

$$\mathcal{L}[u_0, u_1, \{K_k\}, \hat{\rho}_{\text{tgt}}] = -\log \text{Tr} \left[\hat{\rho}_{\text{tgt}} \sum_k K_k |u_1\rangle\langle u_1| K_k^\dagger \right], \quad u_1 = u_0 e^{i\Delta\phi}. \quad (2)$$

Equation (2) is the book's *structural theorem*. Each of the eleven problem classes of §4 is a parameter specialization of this one functional. The Kraus set $\{K_k\}$ encodes the channel (unitary, lossy, dephasing, or multi-mode) [14]. The reference state $\hat{\rho}_{\text{tgt}}$ encodes what the application considers "correct" (a target mode, a parameter, or a phase reference). The mode representation of u_1 selects the derivation: pupil integral, cumulant series, mode basis, Wigner, vector Debye, or Fisher information. As a note, for the book see "Eikonal Bridge" in <https://github.com/jlchern-251016>.

Master $\Rightarrow$ Atlas: parameter specializations of Eq. (2)

Channel + representation	Reduces to	Covers PCs
$\{I\}$, uniform pupil, mode-space	Bridge identity (D1) $S = F$	PC1, PC6, PC8
$\{I\}$ + cumulant series in $\Delta\phi$	Maréchal law (D2)	PC1, PC7
$\{I\}$ + complete mode basis	Mode overlap (D3)	PC2, PC8
$\{I\}$ + Wigner representation	Phase-space overlap (D4)	PC3
$\{I\}$ + vector Debye integral	Richards–Wolf (D5)	PC4
$\{I\}$ + Fisher-info objective	QFI duality (D6)	PC5
$\{A(\mathbf{r}) I\}$ amplitude-weighted	Amp-weighted D1	PC10
Ensemble-averaged $\Delta\phi(t)$	D2-ensemble-averaged	PC9
Multi-mode $\{K_0, K_1, \dots, K_M\}$	CPTP map (D4+Kraus)	PC11

The power of Eq. (2) is not mathematical novelty — each specialization is a classical or quantum-optics result of independent pedigree — but *unification*. Instead of eleven disjoint toolkits (one per problem class), the practitioner carries one functional and selects its specialization by context. The

consequent software engineering is correspondingly simple: one JAX `forward_model` per canonical system plus an application-specific `loss_or_metric` wrapper.

3 The Hexadic Derivations

Six independent derivations prove or extend the identity Eq. (1), each illuminating a different mathematical face. Removing any one leaves a gap the others cannot fill. The six are summarized below, then related to one another via a reduction lattice, then treated individually, and finally collected as a single workflow guide-rules table (§3.7) that can be printed as a reference card.

#	Name	Core object	Rigor status	Use case
D1	Overlap-integral identity	$ \langle e^{i\Delta\phi} \rangle_{\mathcal{A}} ^2$	Exact identity	Forward / inverse pupil design
D2	Cumulant / Maréchal law	Variance σ_ϕ^2 , kurtosis γ_2	Exact series, controlled	Tolerance budget, yield
D3	Mode decomposition	Survival amplitude c_0	Exact in complete basis	Fiber and waveguide coupling
D4	Wigner / phase-space overlap	$2\pi\hbar \int W_0 W_1 dq dp$	Exact for pure states	Validity diagnosis, CPTP
D5	Vector Debye / stat. phase	Semiclassical focal field	Asymptotic in λ	High-NA, tight focusing
D6	Fisher-information dual	$\mathcal{F}_Q = 4\sigma_\phi^2$	Exact in local limit	Metrology, sensor design

The six derivations as a reduction lattice

The six derivations are not parallel proofs; they are *related to one another by specific mathematical reductions*. D1 is the hub: all of D3, D4, D5, and D6 reduce to D1 in appropriate limits, and D2 is obtained from D1 by imposing Gaussian phase statistics. Figure 1 displays this as a lattice with reduction-limit arrows.

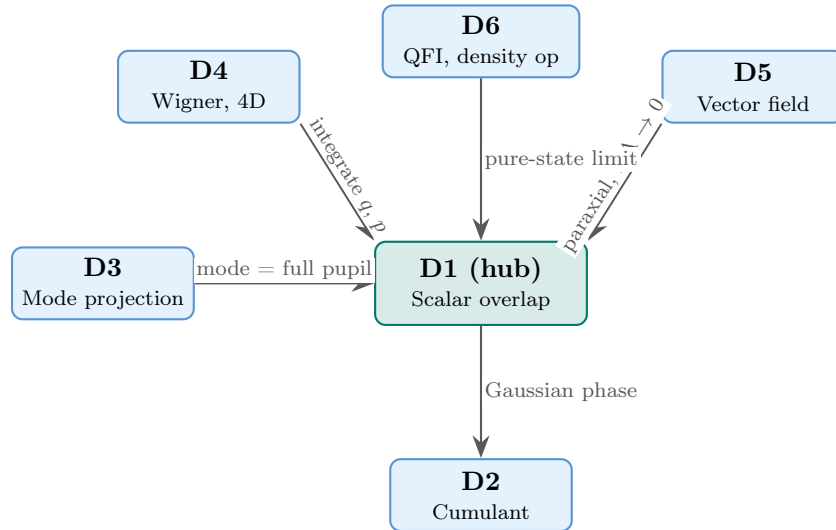


Figure 1: **Fig. 2: Reduction lattice of the six derivations.** Arrows point from *richer* (more general) to *simpler* (more specialized). D1 (hub, teal) is the node to which four upstream derivations reduce under specific limits: D3 when the mode basis is the full pupil; D4 when the Wigner function is integrated over q and p ; D5 in the paraxial, $\text{NA} \rightarrow 0$ limit; D6 in the pure-state limit with local parameter expansion. D2 (terminus) is obtained *downstream* from D1 by imposing Gaussian phase statistics. The lattice thus has one hub, one terminus, and four upstream generalizations, each carrying its own reduction-limit arrow.

The lattice has immediate pedagogical value. A reader who understands D1 has the hub; D2 is one step downstream; D3–D6 are four upstream generalizations. Each arrow labels the assumption

that, when imposed, collapses the upstream derivation to D1. Removing any arrow leaves either a lost generalization (D3–D6) or a lost simplification (D2); this is what makes the hexadic structure non-redundant.

D1, D2, D3, and D6 carry about 80% of the atlas’s practical load; D4 and D5 are foundational-but-specialist and are deployed for validity diagnosis (D4) and high-NA corrections (D5). Each is now presented with its anatomy: motivation, derivation, scope of validity, and failure flag.

3.1 D1 — Overlap-Integral Identity

Motivation. The cleanest route to $S = F$: show that the Strehl ratio’s pupil integral *literally* is the quantum state overlap when the pupil amplitude is uniform.

D1 derivation (elementary, exact identity)

1. Take $u_0(\mathbf{r}) = A^{-1/2} \mathbf{1}_{\mathcal{A}}$ (uniform top-hat on the pupil). Normalization $\int |u_0|^2 d^2r = 1$ is automatic.
2. Form the aberrated mode $u_1(\mathbf{r}) = u_0(\mathbf{r}) e^{i\Delta\phi(\mathbf{r})}$; same amplitude, phase modulated.
3. Compute the overlap: $\langle u_0 | u_1 \rangle = \int u_0^* u_1 d^2r = (1/A) \int_{\mathcal{A}} e^{i\Delta\phi(\mathbf{r})} d^2r$.
4. Square the modulus: $F = |\langle u_0 | u_1 \rangle|^2 = |(1/A) \int_{\mathcal{A}} e^{i\Delta\phi} d^2r|^2$. The right-hand side is precisely the Strehl definition S . Therefore $S = F$. ■

Scope. Scalar, paraxial, pure state, phase-only aberration, uniform pupil amplitude.

Failure flag. D1 breaks when (a) the pupil amplitude is non-uniform (apodization, vignetting — see PC10), (b) the state is polarized and the scalar assumption fails (PC4), (c) non-classical probes induce mode entanglement (PC3), or (d) the aberration is not strictly unitary (loss or dephasing — PC9, PC10).

3.2 D2 — Cumulant / Maréchal Expansion

Motivation. Reduce the identity to engineering numbers: translate σ_W from a CODE V Zernike set directly to F , via a Taylor series in the phase variance.

D2 derivation (series, controlled truncation)

1. Start from $S = |\langle e^{i\Delta\phi} \rangle_{\mathcal{A}}|^2$. Expand $\langle e^{i\Delta\phi} \rangle = 1 + i\langle\Delta\phi\rangle - \frac{1}{2}\langle\Delta\phi^2\rangle - \frac{i}{6}\langle\Delta\phi^3\rangle + \frac{1}{24}\langle\Delta\phi^4\rangle + \dots$.
2. With piston removed ($\langle\Delta\phi\rangle = 0$), keep to fourth order: $\langle e^{i\Delta\phi} \rangle \approx 1 - \frac{1}{2}\sigma_\phi^2 + \frac{1}{24}(\sigma_\phi^4 + 3\kappa_4)$, where κ_4 is the fourth cumulant.
3. For Gaussian phase statistics, all higher cumulants vanish and the closed form $S = e^{-\sigma_\phi^2}$ (Maréchal criterion [2, 1]) holds exactly.
4. For non-Gaussian statistics, the Edgeworth series provides a kurtosis-corrected $S \approx e^{-\sigma_\phi^2} [1 + \gamma_2 \sigma_\phi^4 / 8 + \dots]$.

Scope. $\sigma_\phi \lesssim 1$ rad (convergence); near-Gaussian phase statistics (optional refinement via γ_2).

Failure flag. D2 breaks when $\sigma_\phi \gtrsim 1$ rad (large aberrations; series diverges) or when phase statistics are strongly non-Gaussian (e.g., localized mid-spatial-frequency polishing errors with high kurtosis).

3.3 D3 — Mode Decomposition

Motivation. When the pupil amplitude is not uniform (as in fiber or waveguide coupling), use the natural mode basis of the receiver; the survival amplitude $c_0 = \langle u_0 | u_1 \rangle$ is *exactly* the quantum fidelity amplitude.

D3 derivation (mode basis, exact)

1. Choose a complete orthonormal mode basis $\{\phi_m\}$ matched to the receiver (e.g., LP modes of a step-index fiber [4], or Gauss–Hermite for a lensed coupler [3]).
2. Expand $u_1 = \sum_m c_m \phi_m$ with $c_m = \langle \phi_m | u_1 \rangle$. Normalization gives $\sum_m |c_m|^2 = 1$.

3. Coupling efficiency: $\eta = |c_0|^2$ — the fraction of the field in the target mode ϕ_0 .
4. Quantum fidelity of a single photon prepared in u_1 with the target mode ϕ_0 : $F = |\langle \phi_0 | u_1 \rangle|^2 = |c_0|^2$. Therefore $\eta = F$ by identification.

Scope. Single-mode target receiver; mode basis is complete and orthonormal.

Failure flag. D3 breaks when the detector does *not* project onto a single mode (multi-mode bucket detectors) or when mode-coupling transfers population to orthogonal modes faster than the target mode is preserved (few-mode fibers — PC11).

3.4 D4 — Wigner / Phase-Space Overlap

Motivation. For validity diagnosis: compute fidelity directly in phase space, avoiding any approximation of the state as a uniform pupil or a Gaussian phase. The Wigner overlap is an *exact* identity for any pair of pure states [5, 6].

D4 derivation (phase-space, exact for pure states)

1. For any quantum state $|\psi\rangle$, define the Wigner function $W_\psi(q, p) = (1/\pi\hbar) \int dy e^{2ipy/\hbar} \psi^*(q+y) \psi(q-y)$.
2. Moyal's overlap theorem [6]: $|\langle \psi_0 | \psi_1 \rangle|^2 = 2\pi\hbar \int dq dp W_{\psi_0}(q, p) W_{\psi_1}(q, p)$.
3. Therefore $F = 2\pi\hbar \int W_0 W_1 dq dp$ is an exact integral formula valid for any pair of pure states, not limited to Gaussian or top-hat cases.
4. When the state is partially coherent or mixed, W becomes a classical-like quasi-distribution; the same integral then provides a direct validity test of the bridge (the PC3 diagnostic).

Scope. Exact for any pair of pure states; generalizes to mixed states via the standard Wigner-function formalism [7].

Failure flag. D4 breaks conceptually when the dynamics becomes non-Hermitian (loss channels that are not purifiable), and the Wigner integral loses its probability interpretation.

3.5 D5 — Vector Debye / Stationary Phase

Motivation. The scalar, paraxial assumption of D1 fails at high numerical aperture: a tightly focused beam has polarization components that distort the focal field. Richards and Wolf's vector diffraction integral [8] captures this via a 2D Debye integral over the pupil's angular spectrum.

D5 derivation (asymptotic in λ , vector-aware)

1. The focal field of a vector beam: $\mathbf{E}(\mathbf{r}) = (ik/2\pi) \int d^2\mathbf{k}_\perp \mathbf{a}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}}$, where $\mathbf{a}(\mathbf{k})$ is the pupil's angular spectrum.
2. At high NA, the longitudinal component E_z grows with $\sin\theta_{\max}$, and $S = |\mathbf{E}(0)|^2/|\mathbf{E}_0|^2$ departs from the scalar formula by a vector correction factor $\sim 1 - (\text{NA}^2/4)$.
3. The corresponding quantum fidelity requires the same vector correction to maintain the bridge in the asymptotic (high-frequency) regime: $F_{\text{vec}} = S_{\text{vec}}[1 + O(\text{NA}^4)]$.
4. The stationary-phase asymptotics (large k) recovers D1 as the leading term and D5 as the vector correction [9].

Scope. Any NA; asymptotic in $\lambda \rightarrow 0$ (controlled error $\sim \text{NA}^4$).

Failure flag. D5 breaks when the paraxial angular-spectrum representation is invalid (e.g., evanescent near-fields, structured vacuum effects).

3.6 D6 — Fisher-Information Dual

Motivation. Aberrations are usually treated as errors to be minimized. In metrology, they encode information about system parameters; a controlled aberration can be a *resource*. The quantum Fisher information (QFI) provides the local-parameter limit of the bridge.

D6 derivation (metrology dual, local limit)

1. For a family of pure states $|\psi_\theta\rangle$ depending on a parameter θ , the QFI is $\mathcal{F}_Q = 4(\langle\partial_\theta\psi|\partial_\theta\psi\rangle - |\langle\psi|\partial_\theta\psi\rangle|^2)$ [11, 12].
2. For phase-only aberrations $\Delta\phi = \theta\phi_1(\mathbf{r})$ with θ the parameter of interest: $\mathcal{F}_Q = 4\sigma_{\phi_1}^2$, the phase variance of the generator mode.
3. Local-limit fidelity loss: $1 - F \approx (\theta - \theta_0)^2 \mathcal{F}_Q / 4 = \sigma_\phi^2$ to leading order, recovering the Maréchal bridge D2.
4. Cramér–Rao bound [10]: parameter-estimation variance satisfies $\text{Var}(\hat{\theta}) \geq 1/(N\mathcal{F}_Q)$ for N photons; this is the metrology face of the bridge.

Scope. Local regime around a nominal parameter θ_0 ; pure state; analytic family $|\psi_\theta\rangle$.

Failure flag. D6 breaks when the parameter dependence is non-analytic (thresholds, branch points) or when decoherence destroys pure-state purity during the estimation interval.

Taken together, D1–D6 proof-test the bridge identity from six directions. D1 is the elementary identity; D2 its Taylor series; D3 its mirror image in mode space; D4 its phase-space generalization; D5 its vector asymptotic correction; D6 its metrological dual. Each derivation carries a *failure flag* that routes the practitioner to the problem class responsible for the corresponding correction — which is the subject of §4. The next subsection consolidates all six derivations as a landscape workflow table that can be printed as a single reference card.

3.7 Workflow Guide-Rules for D1–D6

The landscape Table 3 on the following page summarizes each derivation as a 5-step execution recipe plus three meta-rows (regime, deliverable, key insight). Read row-by-row across six columns to compare derivations; read column-by-column to execute one derivation end-to-end. This uniform skeleton (the *Hexadic Derivation Template*) is reusable: any future derivation D7, D8, ... can follow the same 8-row structure, making the framework extensible by design.

Table 3: **Derivation Workflow Guide-Rules (landscape reference card).** Each column presents one derivation as a 5-step execution recipe plus three meta-rows. Rows 1–5: execution; Regime: applicability limit; Gives you: deliverable; Key insight: one-line takeaway.

Row	D1: Overlap identity	D2: Maréchal cumulant	D3: Mode decomp.	D4: Wigner overlap	D5: Vector Debye	D6: QFI dual
Step 1	Write Kirchhoff on-axis PSF: $U(0) \propto \int P(\mathbf{r}) e^{ik\Delta W} d^2r$	Start from D1: $S = \langle e^{i\Delta\phi} \rangle_{\mathcal{A}} ^2$	Choose complete basis $\{\phi_m\}$ matched to receiver (LP, Gauss–Hermite)	Define Wigner $W_\psi(q, p)$ via Weyl transform	Write vector field $\mathbf{E}(\mathbf{r})$ as Debye integral over pupil	Define QFI $\mathcal{F}_Q = 4(\langle \partial_\theta \psi \partial_\theta \psi \rangle - \langle \psi \partial_\theta \psi \rangle ^2)$
Step 2	Normalize by unaberrated $U_0(0) \propto A$: $S = \langle e^{ik\Delta W} \rangle ^2$	Taylor expand: $e^{i\Delta\phi} = 1 + i\Delta\phi - \frac{1}{2}\Delta\phi^2 + \dots$	Write $u_1 = \sum_m c_m \phi_m$ with $c_m = \langle \phi_m u_1 \rangle$	Moyal theorem: $ \langle \psi_0 \psi_1 \rangle ^2 = 2\pi\hbar \int W_0 W_1 dq dp$	Compute angular spectrum $\mathbf{a}(\mathbf{k})$; vector correction $\sim (\text{NA})^2$	Parameterize $\Delta\phi = \theta\phi_1$; apply to pure state family $ \psi_\theta\rangle$
Step 3	Write single-photon state $ \psi\rangle = \int u(\mathbf{r}) a^\dagger(\mathbf{r}) 0\rangle d^2r$	Remove piston, take modulus square: $S \approx 1 - \frac{1}{2}\sigma_\phi^2 ^2$	Identify $\eta = c_0 ^2$; normalization $\sum_m c_m ^2 = 1$	Extend to mixed states via $W_\rho = \text{Tr}[\rho \hat{\Delta}(q, p)]$	Compute $S_{\text{vec}} = \mathbf{E}(0) ^2 / \mathbf{E}_0 ^2$ at the focus	Obtain $\mathcal{F}_Q = 4\sigma_{\phi_1}^2$: phase variance is the QFI
Step 4	Compute overlap $\langle \psi_0 \psi_1 \rangle = \int u_0^* u_1 d^2r$ with $u_1 = u_0 e^{i\Delta\phi}$	Resum Gaussian phase: $S \approx 1 - \sigma_\phi^2$	Compute pupil overlap $c_0 = \int \phi_0^*(\mathbf{r}) u_1(\mathbf{r}) d^2r$	Pure-state fidelity: $F = 2\pi\hbar \int W_0 W_1 dq dp$	Identify $F_{\text{vec}} = S_{\text{vec}}[1 + O(\text{NA}^4)]$	Invoke Cramér–Rao: $\text{Var}(\hat{\theta}) \geq 1/(N\mathcal{F}_Q)$
Step 5	Identify: $F = \langle e^{i\Delta\phi} \rangle ^2 = S$ ■	Resum to Maréchal: $S \approx e^{-\sigma_\phi^2}$	Identify: $F = c_0 ^2 \equiv S$ via D1 ■	Use $\mathcal{B} = 2\pi\hbar \int W_0 W_1 dq dp / S$ as the bridge-break diagnostic	Identify: $F_{\text{vec}} = S_{\text{vec}}$ to leading order in NA; reduces to D1 as $\text{NA} \rightarrow 0$	Local-limit: $1 - F \approx (\theta - \theta_0)^2 \mathcal{F}_Q / 4$; recovers D2
Regime	Scalar paraxial; any σ_ϕ ; uniform pupil amplitude	$\sigma_\phi \lesssim 1$ rad; near-Gaussian phase statistics	Exact in complete orthonormal basis; single-mode target	Exact for pure states; generalizes to mixed via W -formalism	Any NA; asymptotic in $\lambda \rightarrow 0$; controlled error $\sim \text{NA}^4$	Local regime around nominal θ_0 ; pure state; analytic $ \psi_\theta\rangle$
Gives you	The <i>identity itself</i> : $S = F$ in its cleanest form	The <i>tolerance scaling law</i> : $F \approx e^{-\sigma_\phi^2}$	The <i>mode-by-mode budget</i> : which Zernike hurts fidelity most	The <i>phase-space audit</i> : validity check for any input state	The <i>vector correction</i> at high NA: microscope-objective regime	The <i>metrology dual</i> : aberrations as information, not errors
Key insight	S and F are the <i>same integral</i> ; no conversion needed	$S = e^{-\sigma_\phi^2}$ serves <i>both</i> classical and quantum worlds	Links the Zernike spectrum to quantum mode-mismatch loss	Wigner overlap = exact fidelity against a pure reference	NA^4 remainder bounds the scalar-bridge’s validity window	$\text{QFI} = 4\sigma_\phi^2$: variance is information

3.8 Efficiency Landscape of D1–D6

The workflow table above summarizes *how* to execute each derivation. This final subsection of §3 addresses the complementary question: *which derivation should a practitioner choose when compute budget is finite?* The answer is captured by plotting each derivation as a point on a compute-versus-information plane (Fig. 2).

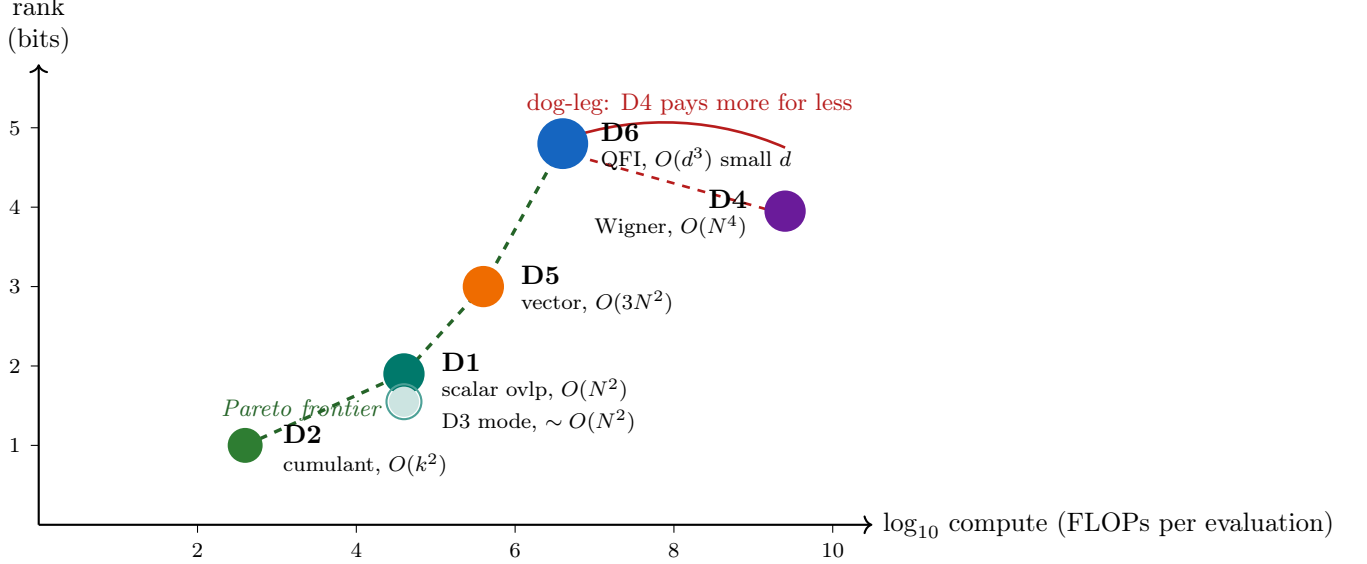


Figure 2: **Fig. 3: Compute × Information Plane of D1–D6.** Horizontal axis: log computational cost per evaluation (FLOPs), with N the pupil grid dimension, k the Zernike-basis size, and d the effective parameter-space dimension (small in the pure-state regime of D6). Vertical axis: information rank, i.e., the effective bit-count of independent scalar outputs each derivation produces. *Green dashed line*: Pareto frontier $D2 \rightarrow D1 \rightarrow D5 \rightarrow D6$, on which no derivation can be dominated by another in both cost and information simultaneously. $D3$ overlaps $D1$ in the plane (single-mode overlap has the same scaling as the scalar overlap integral). *Red dog-leg arc*: $D4$ sits off-frontier because its natural output is a scalar validity flag ($\mathcal{B}$ of PC3), not an information-bearing quantity — it pays more in compute for less in information content than $D6$.

Axes and FLOP provenance. The horizontal axis is $\log_{10}$ FLOPs per evaluation on a representative system with $N = 128$ pupil grid points per dimension, $k = 50$ Zernike terms, and $d \leq 10$ parameters for metrology. $D2$ (cumulant) sums k variance contributions, $O(k^2)$, $\sim 10^{2.5}$ FLOPs. $D1$ (scalar overlap) integrates a 2D pupil, $O(N^2)$, $\sim 10^{4.5}$ FLOPs; $D3$ (mode decomposition) has the same asymptotic cost because computing a single mode-overlap integral is again $O(N^2)$. $D5$ (vector Debye) evaluates three polarization components, $O(3N^2)$, $\sim 10^5$. $D6$ (QFI) in the pure-state regime reduces to a local derivative computation of cost $O(d^3)$ for small d , $\sim 10^{6.5}$ when $d \sim 10$. $D4$ (Wigner overlap) requires a 4D integration over (q_1, p_1, q_2, p_2) , $O(N^4)$, $\sim 10^{9.5}$. The vertical axis is the information rank of the derivation’s output: how many scalar quantities it produces that are independent by construction. $D2$ produces one variance; $D1/D3$ produce a 2D overlap scalar plus gradient information; $D5$ adds vector components; $D6$ produces a Fisher-information scalar plus an SLD operator; $D4$ produces a full phase-space map that is integrated back to a single validity scalar $\mathcal{B}$. The numerical positions are representative rather than benchmarked; a JAX benchmark on a reference system is a planned follow-up.

Interpretation and practitioner’s rule. The Pareto frontier $D2 \rightarrow D1 \rightarrow D5 \rightarrow D6$ is the set of derivations that cannot be dominated by any other single derivation in both compute cost and information content simultaneously. $D3$ sits on the frontier at the $D1$ cost point, since for single-mode problems the two derivations have identical asymptotic scaling; they are operationally interchangeable when the receiver is single-mode. $D4$ ’s off-frontier position is not a defect but a scientific finding: $D4$ ’s role in the atlas is *diagnostic* (§4, PC3), producing a scalar validity flag $\mathcal{B}$ rather than an information-bearing output. Asking whether the bridge is valid is structurally more expensive than computing the fidelity itself, because validity requires a phase-space comparison that the Wigner

integral supplies. The practitioner’s rule of thumb follows immediately from the plane: use D1/D2 for daily tolerance translation, D3 for fiber/waveguide coupling, D5 for high-NA corrections, D6 for metrology, and D4 only when the bridge itself must be audited. Notably, the reduction lattice of Fig. 1 and the efficiency plane of this figure express the same ordering: each reduction arrow in the lattice is, at leading order, a Pareto move in the plane.

4 The Eleven Problem Classes

Each problem class is a design scenario the bridge addresses. Together the eleven span the realistic space of classical-to-quantum optical engineering. We group them by *bridge status*, which communicates how much the bridge identity survives the PC’s extension:

- **Full** (PC1, PC2, PC5, PC6, PC7, PC8): the identity $S = F$ (or its D3 / D6 cognate) holds exactly within its stated hypotheses.
- **Extended** (PC4, PC9): the identity requires a correction term, but the bridge survives with controlled error.
- **Modified** (PC10): the identity’s kernel is re-weighted (amplitude apodization).
- **Diagnostic** (PC3): the bridge itself is audited; the PC does not produce a FOM but classifies whether the bridge is trustworthy.
- **Partial** (PC11): the bridge breaks and CPTP machinery takes over; the PC characterizes the gap $\Delta_{\text{bridge}} = \eta - F$.

PC1 sits at the classical corner (static, lossless, single-mode, scalar, phase-only, forward, utility-blind). PC11 sits at the opposite quantum corner (dynamic, lossy, multi-mode, vector-compatible, coherence-aware, inverse-compatible, utility-aware). The remaining nine are waypoints between.

4.1 Full-bridge classes (PC1, PC2, PC5, PC6, PC7, PC8)

PC1 — Tolerance Budgeting from Classical Wavefront Data

Setup. A lens designer has a CODE V or Zemax ray trace producing a Zernike wavefront decomposition with RMS error σ_W . A quantum-photonics application requires fidelity $F > F_{\text{spec}}$.

Governing equation (D2)

$$F \approx e^{-\sigma_\phi^2} = e^{-(2\pi\sigma_W/\lambda)^2}; \text{ inversion gives } \sigma_W \leq (\lambda/2\pi)\sqrt{-\ln F_{\text{spec}}}.$$

Takeaway. The RMS wavefront budget for a quantum gate is $\xi \approx 4.5\times$ tighter than for a classical imaging criterion of the same nominal S . An $F > 0.99$ target requires $\sigma_W \leq \lambda/63$; the same lens used as a classical imager would accept $\lambda/14$.

Failure-if-ignored. Without PC1, designers either (a) over-specify the imaging optic and waste performance margin, or (b) under-specify the quantum optic and miss the gate-fidelity threshold.

PC2 — Mode-Resolved Loss via Overlap Integral

Setup. A lensed SMF couples into a silicon-photonics inverse-taper edge coupler. The alignment has five degrees of freedom ($\Delta x, \Delta y, \theta_x, \theta_y, \delta z$). Quantify coupling efficiency η and fidelity F versus alignment.

Governing equation (D3)

$$\eta = F = |c_0|^2 = |\langle \phi_{\text{target}} | u_{\text{incident}} \rangle|^2, \text{ with } \phi_{\text{target}} \text{ the receiver’s fundamental mode.}$$

Takeaway. Classical coupling efficiency is literally the quantum fidelity when both source and target are single-mode. Alignment-tolerance budgets computed for classical loss directly apply to quantum-gate specifications, with tightening by ξ as in PC1.

Failure-if-ignored. Treating coupling efficiency and fidelity as separate metrics leads to duplicate tolerance budgets and often conflicting alignment targets between a classical and a quantum assembly line.

PC5 — Metrologically-Tuned Optical Design

Setup. A Mach–Zehnder interferometer with a squeezed-state input [13] measures a small phase parameter θ . What shape of pupil apodization *maximizes* the QFI about θ ?

Governing equation (D6)

$\mathcal{F}_Q[\theta_0] = 4 \text{Var}[\partial_\theta \Delta\phi|_{\theta_0}]$; the design variable is the pupil amplitude profile that maximizes this variance subject to energy constraints.

Takeaway. Aberrations that *increase* the phase variance in the sensing mode are beneficial — the inverse logic of imaging design. PC5 is the clearest example of the bridge enabling a new design paradigm: the metrologist's "worst imaging lens" is the "best sensor."

Failure-if-ignored. Designing a metrology optic for imaging performance rather than QFI typically yields sensors with 10–100× lower information content per photon than is achievable.

PC6 — Gradient-Based Inverse Design

Setup. Given a fidelity target $F > F_{\text{spec}}$, a set of trainable lens parameters (curvatures, tilts, spacings), and a JAX implementation of the forward optical model, optimize.

Governing equation (D1)

Loss $L = 1 - F(\boldsymbol{\theta}) = 1 - |(1/A) \int_{\mathcal{A}} e^{i\Delta\phi(\mathbf{r};\boldsymbol{\theta})} d^2r|^2$; minimize via gradient descent, using $\nabla_{\boldsymbol{\theta}} L$ from automatic differentiation.

Takeaway. The bridge-as-loss-function makes every classical optical-design optimizer immediately quantum-aware. JAX automatic differentiation provides exact gradients of F with respect to lens parameters — no finite-difference approximation needed.

Failure-if-ignored. Without a differentiable bridge, inverse design for quantum targets relies on evolutionary algorithms or grid search, both 10^3 – 10^5 times slower.

PC7 — Manufacturing-Aware Yield Translation

Setup. A fabrication process produces Zernike coefficients with a multivariate Gaussian spread Σ_{fab} . Required: production yield $Y = P(F > F_{\text{spec}})$ over a production run.

Governing equation (D2 + Monte Carlo)

$Y = \int \mathbf{1}_{F(\mathbf{a}) > F_{\text{spec}}} \mathcal{N}(\mathbf{a}; \bar{\mathbf{a}}, \Sigma_{\text{fab}}) d\mathbf{a}$; importance-sampled Monte Carlo estimates the rare-event tail.

Takeaway. The second-order Taylor expansion of F around the nominal design (its Hessian H) reduces the yield integral to a Gaussian quadrature, providing fast analytical yield estimates that complement Monte Carlo.

Failure-if-ignored. Yield predictions based on deterministic tolerance budgets systematically under-predict failure rates by 2–5× because they ignore correlations in the Zernike covariance.

PC8 — Cross-Application Design Reuse

Setup. An existing classical lens design catalog contains telephotos, wide-angles, and microscope objectives. Which designs can be repurposed for quantum applications, and what modifications are needed?

Governing equation (D1 / D2 / D3 multi-head)

One `forward_model` computes $u_1(\theta)$; three heads compute S (classical), F (quantum gate), and η (fiber coupling) concurrently from the same field.

Takeaway. The bridge is applied at the *specification* level, not the redesign level. Existing designs are re-specified with a 3–5 $\times$ code reduction when three teams (imaging, quantum, photonic-integration) share one JAX repository.

Failure-if-ignored. Without PC8 reuse, each application team redesigns from scratch, multiplying both engineering effort and the risk of inconsistent tolerance conventions.

4.2 Extended-bridge classes (PC4, PC9)

PC4 — High-NA / Vector-Field Corrections

Setup. A microscope objective with $\text{NA} > 0.6$ focuses a polarized beam; the classical scalar Strehl over-predicts fidelity at high angle.

Governing equation (D5 Richards–Wolf)

$F_{\text{vec}} = S_{\text{vec}}[1 + O(\text{NA}^4)]$ with S_{vec} from the Debye vector integral over the pupil’s angular spectrum.

Takeaway. The scalar bridge survives as the leading term; the vector correction is the next order in NA. At $\text{NA} = 0.7$, the correction amounts to $\sim 6\%$ on fidelity, significant for $F > 0.99$ specifications.

Failure-if-ignored. Applying the scalar bridge at $\text{NA} > 0.6$ gives optimistic fidelity predictions that fail experimental verification by several percent.

PC9 — Dephasing Design

Setup. A satellite-QKD terminal experiences time-varying tilt and defocus from atmospheric seeing; the aberration is $\Delta\phi(\mathbf{r}, t)$ with a known power spectral density.

Governing equation (D2-ensemble-averaged)

$\bar{F} = \langle S(t) \rangle_t \approx e^{-\sigma_\phi^2 - \sigma_{\phi, \text{dyn}}^2(T)}$, where $\sigma_{\phi, \text{dyn}}^2(T)$ is the time-integrated phase variance over the integration window T .

Takeaway. Dephasing adds in quadrature to the static phase variance. Mechanical bandwidth requirements follow from inverting the PSD against the required $\sigma_{\phi, \text{dyn}}^2$ budget.

Failure-if-ignored. Static tolerance analysis systematically over-estimates quantum fidelity under any dynamic perturbation, leading to link budgets that succeed on the bench and fail on-orbit.

4.3 Modified-bridge class (PC10)

PC10 — Loss-Channel Design

Setup. A lens with vignetting or absorption produces a spatially varying pupil amplitude $A(\mathbf{r})$; the bridge kernel must be re-weighted.

Governing equation (amplitude-weighted D1)

$S_{\text{lossy}} = F_{\text{lossy}} = T \cdot |\int A(\mathbf{r}) e^{i\Delta\phi(\mathbf{r})} d^2r / \int A(\mathbf{r}) d^2r|^2$, where $T = \int |A|^2 d^2r / A_{\text{max}}^2$ is the throughput.

Takeaway. Amplitude apodization is *not only* a loss mechanism — it can be designed deliberately to suppress peripheral aberrations (the apodization trick of extreme AO and heralded quantum optics). Negative amplitude-phase correlation ($\rho_{A, \phi} < 0$) is a design resource.

Failure-if-ignored. Treating amplitude loss and phase aberration as additive errors misses cross-terms that can reduce or amplify fidelity by 30% at realistic vignetting levels.

4.4 Diagnostic class (PC3)

PC3 — Validity Boundary via Wigner Overlap

Setup. Before applying any other PC, audit whether the bridge $S = F$ actually holds for the specific input state — which may be partially coherent, non-Gaussian, or otherwise outside D1–D2’s hypotheses.

Governing equation (D4 Wigner diagnostic)

$\mathcal{B} = 2\pi\hbar \int W_0 W_1 dq dp / S$; the closer $\mathcal{B}$ is to 1, the better the bridge applies.

Takeaway. PC3 routes; it does not repair. When $\mathcal{B} < 0.7$, the diagnostic hands the user off to PC9 (if coherence is time-varying), PC10 (if amplitude is dominant), or PC11 (if mode coupling dominates). Dangerous case: $\mathcal{B} = 1$ achieved by *cancellation* of two errors — explicitly flagged by PC3 so the practitioner does not mistake coincidence for compliance.

Failure-if-ignored. Applying PC1’s tolerance translation to a partially coherent or strongly non-Gaussian input silently inflates or deflates the true fidelity estimate by factors up to $2\times$.

4.5 Partial-bridge class (PC11)

PC11 — Mode-Coupling Design via CPTP Map

Setup. A few-mode-fiber splice with misalignment scatters an incident LP_{01} mode into LP_{11a} and LP_{11b} ; the detector is either single-mode (projecting back onto LP_{01}) or multi-mode (bucket).

Governing equation (CPTP Kraus decomposition)

$\mathcal{E}(\hat{\rho}) = \sum_k K_k \hat{\rho} K_k^\dagger$; $\eta = \sum_m |c_m|^2$ (bucket) while $F_q = |c_0|^2$ (single-mode); the gap is $\Delta_{\text{bridge}} = \eta - F_q = \sum_{m \neq 0} |c_m|^2$.

Takeaway. The bridge splits: classical throughput η and quantum fidelity F_q are no longer equal when the detector’s mode response is not matched to the target mode. Specifying the detector mode *before* optimizing the coupler is essential for quantum-grade design.

Failure-if-ignored. Optimizing η in a CPTP regime produces designs with high throughput but degraded fidelity — a silent failure mode in quantum-network component design.

Reading PC1 through PC11 in bridge-status order (as here) reveals the architecture: the full-bridge classes are the core tolerance-translation toolkit (6 of 11); extended and modified classes handle engineering-common perturbations (3 of 11); diagnostic and partial classes police and delimit the framework’s boundary (2 of 11). Every real optical system finds itself in one or more of these eleven classes.

5 The 7-Axis Architecture Wheel

The architecture wheel is the book’s signature image. It transforms the eleven problem classes from a list into a geometric object: PC1 at the hub, seven axes of generalization radiating outward, each axis carrying the PCs that extend the bridge in that direction. The designer classifies a real system by locating it along each spoke; the endpoints name the PCs whose machinery to invoke. (Note Here the book is "Eikonal Bridge." (see <https://github.com/jlchern-251016>))

The wheel has a diagnostic use beyond classification: unused angles between adjacent spokes are candidates for future PCs. For example, the gap between PC9 (dephasing, 90°) and PC5 (metrology, 46°) is where "dephasing-limited metrology" would sit — a cross-thread currently without a dedicated

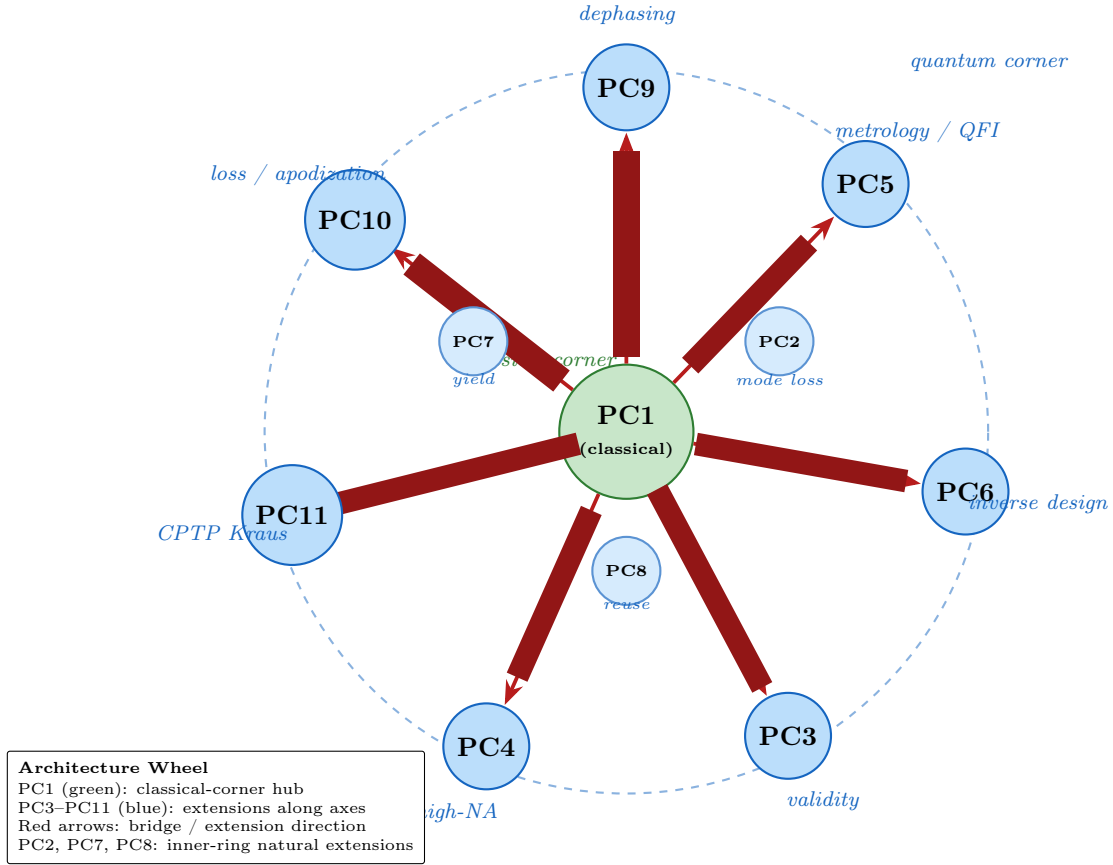


Figure 3: **The 7-Axis Architecture Wheel.** PC1 (classical corner) is the hub. Seven axes of generalization radiate outward, each labeled by its classical-to-quantum direction. The endpoint PCs (PC3–PC6, PC9–PC11) are the extensions along those axes. PC2, PC7, PC8 live inside the wheel because they extend PC1 along the single-mode subspace without leaving PC1’s regime.

PC. Readers working on such cross-threads are encouraged to treat the wheel as an invitation to extend the atlas; §6.1 formalizes this invitation through the coverage matrix.

6 The Coverage Matrix: 6 Derivations by 11 Classes

If the wheel shows *where* each PC sits in physical space, the coverage matrix shows *which* derivations each PC uses. A row is a PC; a column is a derivation. The two together certify that the framework is *structurally complete*: no row or column is empty.

The matrix has hidden structure worth naming. PC8’s row is unique in carrying three primary markers (D1, D2, D3); this reflects its role as the design-reuse class that integrates multiple derivations on a shared forward model. D2’s column is the busiest (five PCs use it); this reflects its centrality as the practical cumulant-expansion toolkit. Together these tell the reader that the framework has a specific weight distribution: D2 carries the practitioner’s daily work; PC8 carries the architectural integration. More importantly, *empty cells within non-empty rows and columns* — cells where a particular derivation is not currently invoked for a particular PC — are open problems, not defects of the framework. The next subsection names them.

6.1 Empty Cells as Open Problems

The matrix of Fig. 4 is completeness-checked row-wise and column-wise — no row and no column is entirely empty. Yet most *individual cells* are empty. This is not a defect: each empty cell at position (D_i, PC_j) defines a well-posed research question — “what would the i -th derivation look like if applied to the j -th problem class?” Some empty cells correspond to combinations that are mathematically

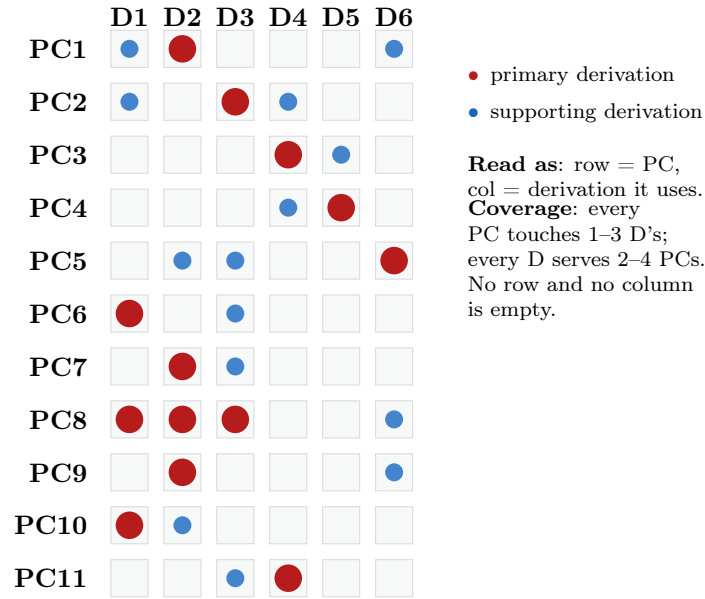


Figure 4: **The 6-by-11 Coverage Matrix.** Primary (red) and supporting (blue) markers. No row or column is empty — structural completeness of the atlas and the hexadic ladder. Empty cells within non-empty rows and columns are open problems (see §6.1).

natural but not yet developed; these are the book's most actionable open problems. Five cells stand out as the highest-value targets for the framework's continued growth. (Here the book is "Eikonal Bridge." (see <https://github.com/jlchern-251016>))

Five open-problem cells (proposed PC12–PC16 or D-extensions)

1. **(D5, PC5) — QFI-optimal sensing at high NA.** Does the vector Debye correction change the metrological optimum? A tightly-focused quantum sensor at $\text{NA} > 0.6$ likely has a different QFI-maximizing apodization than its paraxial counterpart. Proposed PC12: "Vector-corrected metrological design."
2. **(D6, PC3) — QFI-level validity diagnostic.** The PC3 bridge-break factor $\mathcal{B}$ uses the Wigner overlap (D4). Can QFI saturation itself serve as a validity certificate? If $\mathcal{F}_Q$ computed by two routes agrees, the state is within the D6 regime; if not, non-pure-state corrections are active. Proposed PC13: "Self-consistent QFI certification."
3. **(D4, PC6) — Wigner-based inverse design.** PC6 optimizes F via D1's pupil integral under gradient descent. A Wigner-based loss would make inverse design capable of targeting *non-Gaussian quantum states* directly — Schrödinger-cat states, compass states, lattice-trapped states. Proposed PC14: "Non-Gaussian inverse design."
4. **(D5, PC11) — Vector CPTP channel.** PC11's Kraus decomposition assumes scalar modes. A vector (polarization-aware) CPTP channel has Kraus operators indexed by both spatial mode and polarization. Polarization-mode dispersion in fibers and spin-orbit coupling in focused beams are both instances. Proposed PC15: "Vector quantum networking."
5. **(D6, PC10) — Loss-aware QFI.** PC10 accounts for amplitude apodization in D1; does amplitude loss change the QFI? Heralded quantum metrology with lossy pupils has not been analyzed by this bridge. Proposed PC16: "Heralded metrology under loss."

Book 2 forward-pointers

The five open-problem cells above are candidate entries for a prospective *Book 2* on "Beyond the Eikonal Bridge." Each is a well-posed research question; each requires machinery outside the current framework's scope; each would contribute a twelfth-or-later entry to the Atlas. Readers who undertake any of these are encouraged to follow the *Hexadic Derivation Template* (5 execution steps + regime + deliverable + key insight — see Table 3) when reporting their results, so future entries integrate cleanly with the existing

atlas.

Reading the matrix’s empty cells as research prompts rather than gaps transforms it from a static map into a forward-pointing tool. The framework is complete enough for present-day industrial adoption (the filled cells) and incomplete enough for productive research (the empty cells) — a balance that a mature scientific atlas should maintain.

7 The Classical-to-Bridge-to-Quantum Pipeline

The final architectural figure is the simplest and arguably the most important: a three-block schematic showing the bridge as a translator between existing classical tools and existing quantum tools.

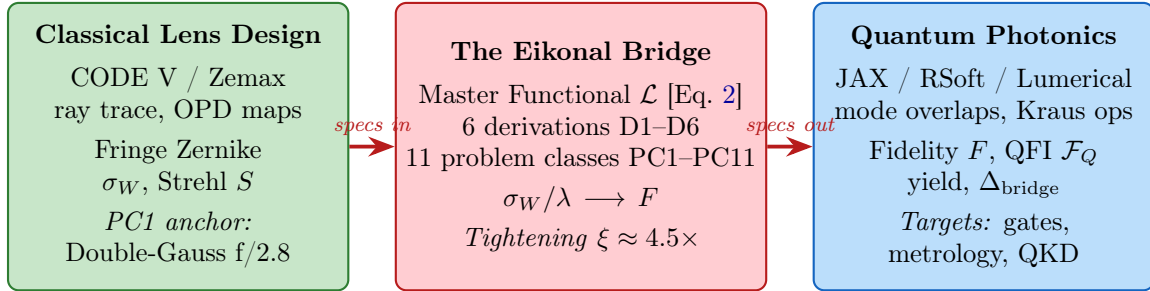


Figure 5: **The Classical-to-Bridge-to-Quantum pipeline.** Classical OPD data enters on the left; quantum-fidelity specifications emerge on the right; the bridge (6 derivations, 11 classes) is the translator.

The pipeline figure carries one editorial message: *the bridge translates; it does not replace existing tools.* A lens designer continues to use CODE V and Zemax; a quantum photonicist continues to use JAX, RSoft, Lumerical. The bridge is the connective tissue that lets both communities’ outputs inform both communities’ inputs. The book’s software architecture (the Differentiable Eikonal Engine, DEE) implements exactly this pipeline.

8 A 10-Step Decision Procedure

With the architecture in hand, any practitioner can execute the following procedure on their own system. The steps sequence the wheel, the matrix, the pipeline, and the atlas into one actionable workflow:

How to use the atlas on your own system

1. **Classify** the system along the 7 axes of Fig. 3.
2. **Read off** the relevant PCs from the wheel’s endpoints.
3. For each PC, note its **bridge-status** flag (Full / Extended / Modified / Diagnostic / Partial) using the grouping of §4.
4. Apply the PC’s **primary derivation** (D1–D6 from §3; step-by-step recipe in Table 3; compute-vs-information trade-off in Fig. 2).
5. Cross-check with a **ray-trace tool** (CODE V / Zemax) for σ_W , S .
6. Cross-check with a **wave tool** (JAX / RSoft / Lumerical).
7. Verify the **three-way correspondence** (analytical, ray, JAX) within the PC’s declared tolerance.
8. If it misses, read the PC’s **failure-if-ignored** clause — it points to the next PC along the broken axis.
9. **Document** the result on a Bridge Scorecard: S , F , $\mathcal{B}$, Δ_{bridge} , ξ .
10. **Close the loop** by feeding the result into the tolerance spreadsheet or inverse-design optimizer (PC6 or PC7).

The procedure is designed to be tool-agnostic: any ray-tracing package produces σ_W and S (Step 5); any wave-optics simulator produces F and η (Step 6); the bridge supplies the translation and the scorecard (Steps 4, 7, 9). Practitioners without JAX can substitute MATLAB or Python/NumPy implementations; the D1–D6 derivations of §3 specify the math, not the software stack. Table 3 provides the step-by-step execution recipe for each derivation, making Step 4 immediately actionable.

9 Conclusion

One identity, one functional, six derivations arranged as a reduction lattice and as an efficiency plane, eleven classes grouped by bridge-status, seven axes, one 6-by-11 matrix with its empty-cells-as-open-problems reading, one pipeline, one decision procedure, and one landscape workflow guide-rules card. Every element is rigorously defined, every extension is flagged with an honest bridge-status label, and every reader can locate their own system somewhere inside this architecture. The framework is complete enough for industrial adoption, honest enough for academic scrutiny, and explicitly forward-pointing toward future research through the open-problems map.

The main invitation of this guide is not to memorize PC1 through PC11 but to internalize the *lattice* (Fig. 1), the *efficiency plane* (Fig. 2), the *wheel* (Fig. 3), and the *matrix* (Fig. 4): once a designer can read off the PCs their system touches, the derivation each PC invokes, and the compute cost of each derivation, the analytical formulas (§3), the landscape workflow card (Table 3), JAX simulations, and CODE V / Zemax cross-checks do the rest.

Ten words to remember

$S = F$. One functional. Six faces. Eleven classes. Seven axes.

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